

Nonlinear Inverse Problems

Synopsis

The nonlinear regression approaches of Chapter 9 are generalized to problems requiring regularization. The Tikhonov regularization and Occam's inversion approaches are introduced. Seismic tomography and electrical conductivity inversion examples are used to illustrate the application of these methods. Resolution analysis for nonlinear problems is addressed.

10.1. REGULARIZING NONLINEAR LEAST SQUARES PROBLEMS

As with linear problems, the nonlinear least squares approaches can run into difficulty with ill-conditioned problems. This typically happens as the number of model parameters grows. Here, we will discuss regularization of nonlinear inverse problems and algorithms for computing a regularized solution to a nonlinear inverse problem.

The basic ideas of Tikhonov regularization can be extended to nonlinear problems. Suppose that we are given a nonlinear discrete inverse problem where an n -element model $\mathbf{m}$ and an m -element data vector $\mathbf{d}$ are related by a nonlinear system of equations $\mathbf{G}(\mathbf{m}) = \mathbf{d}$. For convenience, we will assume that the m nonlinear constraint equations have been scaled to incorporate the measurement standard deviations σ_i . We seek the solution with the smallest seminorm $\|\mathbf{Lm}\|_2$ that comes sufficiently close to fitting the data vector, where $\mathbf{L}$ is an appropriate roughening matrix (e.g., (4.28)).

We can formulate this problem as

$$\begin{aligned} \min \quad & \|\mathbf{Lm}\|_2 \\ & \|\mathbf{G}(\mathbf{m}) - \mathbf{d}\|_2 \leq \delta. \end{aligned} \quad (10.1)$$

Note that the form of the problem is virtually identical to that which was considered in the linear case (e.g., (4.25)), with the only difference being that we now have a general function $\mathbf{G}(\mathbf{m})$ instead of a matrix-vector multiplication $\mathbf{Gm}$. As in the linear case, we can reformulate this problem in terms of minimizing the misfit subject to a constraint on $\|\mathbf{Lm}\|_2$,

$$\begin{aligned} \min \quad & \|\mathbf{G}(\mathbf{m}) - \mathbf{d}\|_2 \\ & \|\mathbf{Lm}\|_2 \leq \epsilon, \end{aligned} \quad (10.2)$$

or as a regularized (often referred to as “damped”) least squares problem,

$$\min \|\mathbf{G}(\mathbf{m}) - \mathbf{d}\|_2^2 + \alpha^2 \|\mathbf{Lm}\|_2^2. \quad (10.3)$$

All three versions of the regularized least squares problem can be solved by applying standard nonlinear optimization software. In particular, (10.3) is a nonlinear least squares problem, so we can apply the LM or GN methods to it. Of course, any such approach will still have to deal with the possibility of local minima that are not global minimum points. In some cases, it is possible to show that nonlinear least squares problems are convex, and thus possess only global minima. In other cases we will have to employ multistart or some other global optimization strategy to determine whether there are multiple minima.

To apply the GN method to (10.3), we rewrite it as

$$\min \left\| \begin{bmatrix} \mathbf{G}(\mathbf{m}) - \mathbf{d} \\ \alpha \mathbf{Lm} \end{bmatrix} \right\|_2^2. \quad (10.4)$$

The Jacobian of (10.4) for the k th iteration is

$$\mathbf{K}(\mathbf{m}^k) = \begin{bmatrix} \mathbf{J}(\mathbf{m}^k) \\ \alpha \mathbf{L} \end{bmatrix}, \quad (10.5)$$

where $\mathbf{J}(\mathbf{m}^k)$ is the Jacobian of $\mathbf{G}(\mathbf{m}^k)$. A GN model step is obtained by applying (9.29) and solving

$$\mathbf{K}(\mathbf{m}^k)^T \mathbf{K}(\mathbf{m}^k) \Delta \mathbf{m} = -\mathbf{K}(\mathbf{m}^k)^T \begin{bmatrix} \mathbf{G}(\mathbf{m}^k) - \mathbf{d} \\ \alpha \mathbf{Lm}^k \end{bmatrix}, \quad (10.6)$$

or, combining (10.5) and (10.6), by solving

$$\left(\mathbf{J}(\mathbf{m}^k)^T \mathbf{J}(\mathbf{m}^k) + \alpha^2 \mathbf{L}^T \mathbf{L} \right) \Delta \mathbf{m} = -\mathbf{J}(\mathbf{m}^k)^T (\mathbf{G}(\mathbf{m}^k) - \mathbf{d}) - \alpha^2 \mathbf{L}^T \mathbf{Lm}^k. \quad (10.7)$$

Equation (10.7) resembles the LM method (9.30). Note, however, that α in (10.7) now appears in the objective function being minimized (10.4) and thus introduces regularization. To further stabilize the iterations, as in the LM method, a variable $\lambda \mathbf{I}$ term could be added to the matrix term of the left-hand side of (10.7) to steer iterative updates toward the direction of steepest descent. This will not be necessary if the explicit regularization of (10.7) sufficiently stabilizes the system of equations.

Example 10.1

Consider a modified version of the cross-well tomography example from Exercise 4.3. We introduce nonlinearity by employing a more realistic forward model that incorporates ray-path bending due to seismic velocity changes. The two-dimensional velocity



structure is parameterized using a 64-element matrix of uniformly spaced slowness nodes on an 8 by 8 grid spanning a 1600 m by 1600 m region.

We apply an approximate ray-bending technique to estimate refracted ray paths within the slowness model and to estimate travel times and their partial derivatives with respect to the model parameters [167]. Figure 10.1 shows the true velocity model and a corresponding set of 64 ray paths. The true model consists of a background velocity of 2.9 km/s with large embedded fast (+10%) and slow (−15%) Gaussian-shaped anomalies. The data set consists of the 64 travel times between each pair of opposing sources and receivers with $N(0, (0.001 \text{ s})^2)$ noise added.

Note that refracted ray paths tend to be “repelled” from low-velocity regions (dark shading) and are, conversely, “attracted” to high-velocity regions (light shading) in accordance with Fermat’s least-time principle. In practice this effect makes low-velocity regions more difficult to resolve in such studies because they will be less well sampled by ray paths.

A discrete approximation of the two-dimensional Laplacian operator is used to regularize this problem. Iterative GN (10.7) solutions were obtained for a range of 16 values of α ranging logarithmically from approximately 4.9 to 367. Figure 10.2 shows the suite of solutions after five iterations. An L-curve plot of seminorm versus data misfit is plotted in Figure 10.3, along with the discrepancy principle value $\delta = 0.001 \cdot \sqrt{64} = 0.008$. Note that the two most lightly regularized solutions are out of their expected monotonically decreasing residual norm positions on the L-curve. This is because the linearization-based GN method is unable to accurately solve the corresponding poorly conditioned least squares systems (10.7) in these cases. Such solutions

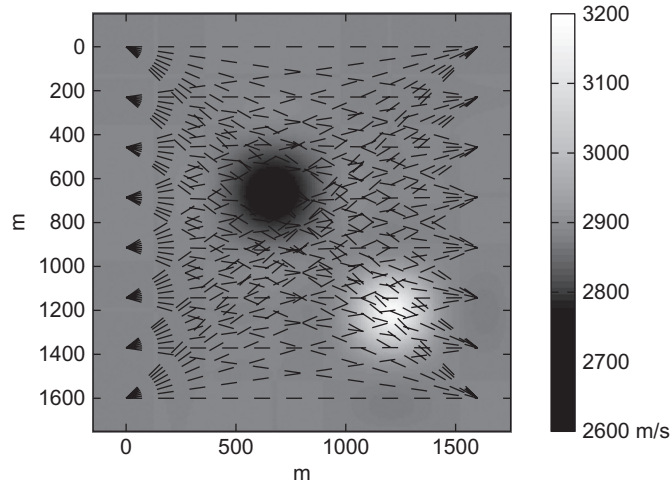


Figure 10.1 True velocity model and the corresponding ray paths for the bent-ray cross-well tomography example.

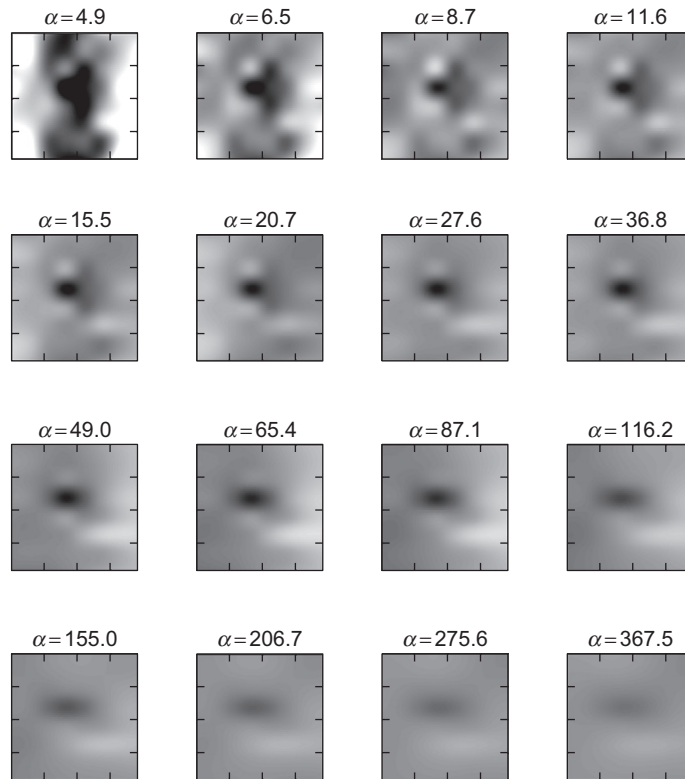


Figure 10.2 Suite of GN (10.7), second-order regularized solutions, ranging from least (upper left) to most (lower right) regularized, and associated α values. The physical dimensions and gray scale are identical to those of Figures 10.1 and 10.4.

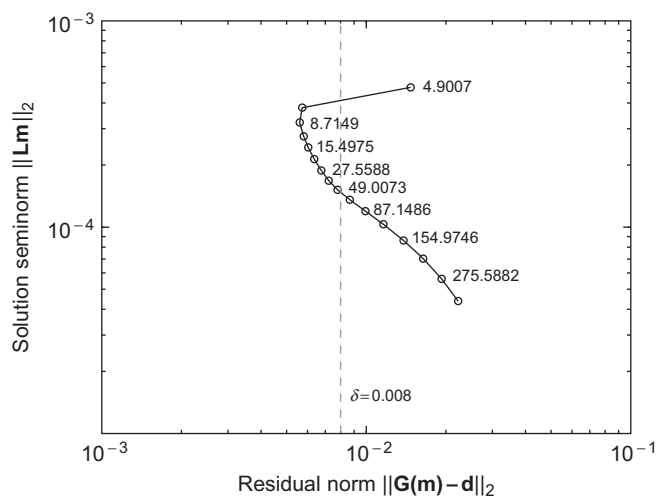


Figure 10.3 L-curve and corresponding α values for the solutions of Figure 10.2.

could be improved by modifying the system of equations with an LM stabilizing term as described above. The solution best satisfying the discrepancy principle corresponds to $\alpha \approx 37$ (Figure 10.4).

Because we know the true model in this example, it is instructive to examine how well the regularized solutions of Figure 10.2 compare to it. Figure 10.5 shows the 2-norm model misfit as a function of α , and demonstrates that the discrepancy principle solution for this problem, and for this particular noise realization, is indeed close to

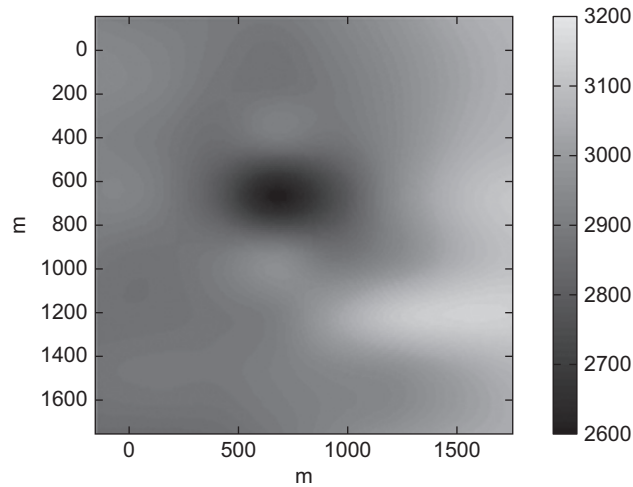


Figure 10.4 Best-solution velocity structure (m/s), α selected using the discrepancy principle, $\alpha \approx 37$.

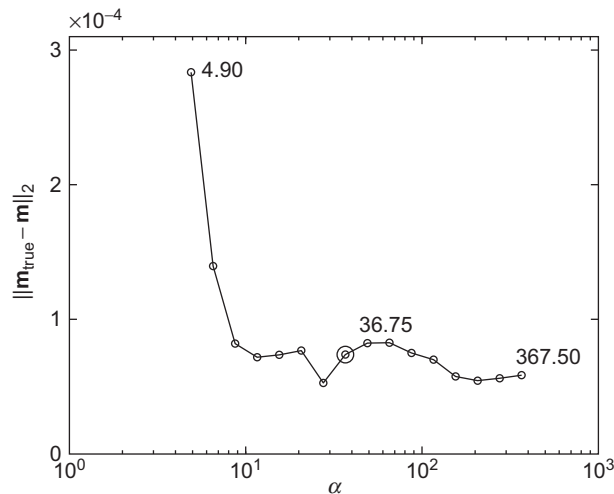


Figure 10.5 Model misfit 2-norm as a function of regularization parameter α , with preferred model highlighted.

the minimum in $\|\mathbf{m} - \mathbf{m}_{\text{true}}\|_2$. Note that the solution shown in Figure 10.4 exhibits resolution artifacts that are common in regularized solutions, such as streaking, side lobes, and amplitude under-recovery (see Example 10.3).

10.2. OCCAM'S INVERSION

Occam's inversion is a popular algorithm for nonlinear inversion introduced by Constable, Parker, and Constable [33]. The name refers to the 14th-century philosopher William of Ockham, who argued that simpler explanations should always be preferred to more complicated explanations. A similar statement occurs as rule 1 in Newton's "Rules for the Study of Natural Philosophy" [118]. This principle has become known as "Occam's razor."

Occam's inversion uses the discrepancy principle, and searches for the solution that minimizes $\|\mathbf{Lm}\|_2$ subject to the constraint $\|\mathbf{G}(\mathbf{m}) - \mathbf{d}\|_2 \leq \delta$. The algorithm is straightforward to implement, requires only the nonlinear forward model $\mathbf{G}(\mathbf{m})$ and its Jacobian, and works well in practice.

We assume that our nonlinear inverse problem has been cast in the form of (10.1). The roughening matrix $\mathbf{L}$ can be $\mathbf{I}$ to implement zeroth-order Tikhonov regularization, or it can be a finite difference approximation of a first (4.27) or second (4.28) derivative for higher-order regularization. In practice, Occam's inversion is often used on two- or three-dimensional problems where $\mathbf{L}$ is a discrete approximation of the Laplacian operator.

As usual, we will assume that the measurement errors in $\mathbf{d}$ are independent and normally distributed. For convenience, we will also assume that the system of equations $\mathbf{G}(\mathbf{m}) = \mathbf{d}$ has been scaled so that the standard deviations σ_i are equal.

The basic idea behind Occam's inversion is an iteratively applied local linearization. Given a trial model $\mathbf{m}^k$, Taylor's theorem is applied to obtain the local approximation,

$$\mathbf{G}(\mathbf{m}^k + \Delta\mathbf{m}) \approx \mathbf{G}(\mathbf{m}^k) + \mathbf{J}(\mathbf{m}^k)\Delta\mathbf{m}, \quad (10.8)$$

where $\mathbf{J}(\mathbf{m}^k)$ is the Jacobian,

$$\mathbf{J}(\mathbf{m}^k) = \begin{bmatrix} \frac{\partial G_1(\mathbf{m}^k)}{\partial m_1} & \cdots & \frac{\partial G_1(\mathbf{m}^k)}{\partial m_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial G_m(\mathbf{m}^k)}{\partial m_1} & \cdots & \frac{\partial G_m(\mathbf{m}^k)}{\partial m_n} \end{bmatrix}. \quad (10.9)$$

Using (10.8), the regularized least squares problem (10.3) becomes

$$\min \|\mathbf{G}(\mathbf{m}^k) + \mathbf{J}(\mathbf{m}^k)\Delta\mathbf{m} - \mathbf{d}\|_2^2 + \alpha^2 \|\mathbf{L}(\mathbf{m}^k + \Delta\mathbf{m})\|_2^2, \quad (10.10)$$

where the variable is $\Delta \mathbf{m}$ and $\mathbf{m}^k$ is constant. Reformulating this as a problem in which the variable is $\mathbf{m}^{k+1} = \mathbf{m}^k + \Delta \mathbf{m}$, and letting

$$\hat{\mathbf{d}}(\mathbf{m}^k) = \mathbf{d} - \mathbf{G}(\mathbf{m}^k) + \mathbf{J}(\mathbf{m}^k)\mathbf{m}^k, \quad (10.11)$$

gives

$$\min \|\mathbf{J}(\mathbf{m}^k)(\mathbf{m}^k + \Delta \mathbf{m}) - (\mathbf{d} - \mathbf{G}(\mathbf{m}^k) + \mathbf{J}(\mathbf{m}^k)\mathbf{m}^k)\|_2^2 + \alpha^2 \|\mathbf{L}(\mathbf{m}^k + \Delta \mathbf{m})\|_2^2 \quad (10.12)$$

or

$$\min \|\mathbf{J}(\mathbf{m}^k)\mathbf{m}^{k+1} - \hat{\mathbf{d}}(\mathbf{m}^k)\|_2^2 + \alpha^2 \|\mathbf{L}(\mathbf{m}^{k+1})\|_2^2. \quad (10.13)$$

Because $\mathbf{J}(\mathbf{m}^k)$ and $\hat{\mathbf{d}}(\mathbf{m}^k)$ are constant within a given iteration, (10.13) is in the form of a regularized linear least squares problem. If the system is of full rank, the solution is given by

$$\mathbf{m}^{k+1} = \mathbf{m}^k + \Delta \mathbf{m} = \left(\mathbf{J}(\mathbf{m}^k)^T \mathbf{J}(\mathbf{m}^k) + \alpha^2 \mathbf{L}^T \mathbf{L} \right)^{-1} \mathbf{J}(\mathbf{m}^k)^T \hat{\mathbf{d}}(\mathbf{m}^k). \quad (10.14)$$

Note that this method is similar to the GN method applied to the regularized least squares problem (10.3) (see Exercise 10.1). The difference is that in Occam's inversion the parameter α is dynamically adjusted so that the solution will not exceed the allowable misfit. At each iteration we pick the largest value of α that keeps the χ^2 value of the solution from exceeding the bound on δ specified in (10.1). If this is impossible, we instead pick the value of α that minimizes the χ^2 value. At the end of the procedure, we should have a solution with $\chi^2 = \delta^2$. We can now state the algorithm.

Algorithm 10.1 Occam's Inversion

Beginning with an initial solution, $\mathbf{m}^0$, repeat the following steps to compute a sequence of solutions $\mathbf{m}^k$. Stop if and when the sequence converges to a solution with $\chi^2 = \delta^2$.

1. Calculate the Jacobian $\mathbf{J}(\mathbf{m}^k)$ and the vector $\hat{\mathbf{d}}(\mathbf{m}^k)$.
2. Calculate updated models corresponding to a range of regularization parameter values, α ,

$$\mathbf{m}^{k+1} = \left(\mathbf{J}(\mathbf{m}^k)^T \mathbf{J}(\mathbf{m}^k) + \alpha^2 \mathbf{L}^T \mathbf{L} \right)^{-1} \mathbf{J}(\mathbf{m}^k)^T \hat{\mathbf{d}}(\mathbf{m}^k). \quad (10.15)$$

3. Choose the particular $\mathbf{m}^{k+1}$ with the largest value of α such that $\chi^2(\mathbf{m}^{k+1}) \leq \delta^2$. If no such value exists, then use the value of α that minimizes $\chi^2(\mathbf{m}^{k+1})$.
 4. Let $k = k + 1$.
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Example 10.2

We will consider the problem of estimating subsurface electrical conductivities from above-ground electromagnetic induction measurements. The instrument used in this example is the Geonics EM-38 ground conductivity meter. A description of the instrument and the mathematical model of its response can be found in [69]. The mathematical model is complicated, but we will treat it as a black box, and concentrate on the inverse problem.

Measurements are taken at heights of 0, 10, 20, 30, 40, 50, 75, 100, and 150 cm above the surface, with the coils oriented in both the vertical and horizontal orientations. There are a total of 18 observations (Table 10.1). We will assume measurement standard deviations of 0.1 mS/m.

We discretize the subsurface electrical conductivity profile into 10- and 20-cm thick layers and a semi-infinite layer below 2 m, giving us 11 parameters to estimate. The forward problem function $\mathbf{G}(\mathbf{m})$ is available to us as a subroutine. Since we do not have simple formulas for $\mathbf{G}(\mathbf{m})$, we cannot write down analytic expressions for the elements of the Jacobian. However, we can use finite difference approximations to estimate the necessary partial derivatives.

We first apply the LM method to estimate unregularized model parameters. After 50 iterations, the LM method produced the model shown in Figure 10.6. The χ^2 value for this model is 9.62 and there are $18 - 11 = 7$ degrees of freedom, so the model actually fits the data reasonably well. However, the least squares problem is very badly conditioned, with a condition number for $\mathbf{J}^T \mathbf{J}$ of approximately 2×10^{17} . Furthermore, the resulting model is unrealistic because it includes negative electrical conductivities and exhibits the high amplitudes and high-frequency oscillations that are characteristic of under-regularized solutions to inverse problems.

Table 10.1 Data for the EM-38 Example

Height (cm)	EMV (mS/m)	EMH (mS/m)
0	134.5	117.4
10	129.0	97.7
20	120.5	81.7
30	110.5	69.2
40	100.5	59.6
50	90.8	51.8
75	70.9	38.2
100	56.8	29.8
150	38.5	19.9

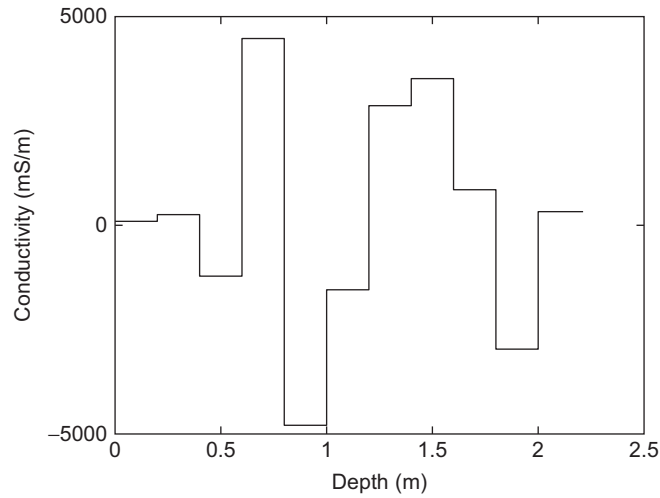


Figure 10.6 LM solution.

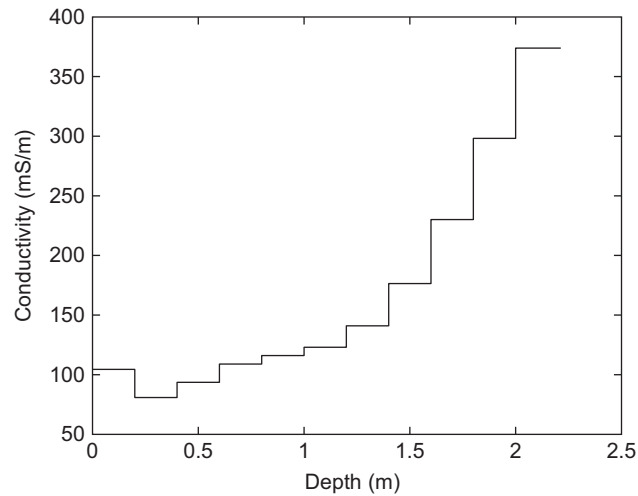


Figure 10.7 Occam's inversion solution.

We next apply Occam's inversion with second-order regularization and a discrepancy principle value of $\delta = 0.1 \cdot \sqrt{18} \approx 0.424$. The resulting model is shown in [Figure 10.7](#), and [Figure 10.8](#) shows the true model. The Occam's inversion method provides a fairly good reproduction of the true model.

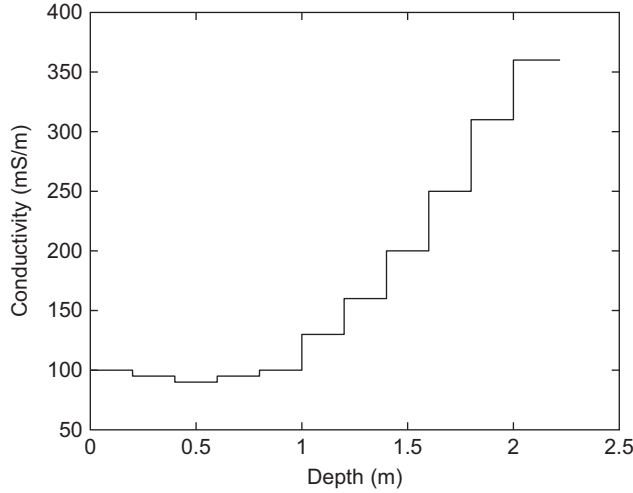


Figure 10.8 True model.

10.3. MODEL RESOLUTION IN NONLINEAR INVERSE PROBLEMS

We introduced the concept of model resolution in Chapter 3 (see (3.63)) by expressing the generalized inverse solution for a linear problem as

$$\mathbf{m} = \mathbf{R}_m \mathbf{m}_{\text{true}} = \mathbf{G}^\dagger \mathbf{G} \mathbf{m}_{\text{true}} \quad (10.16)$$

where $\mathbf{G}$ is a (forward problem) matrix that produces data from a model, $\mathbf{G}^\dagger$ is the generalized inverse matrix used to recover a model from data, and $\mathbf{m}$ and $\mathbf{m}_{\text{true}}$ are the recovered and true models, respectively. In a linear problem, the action of mapping a model to data and then back to a recovered model can be completely characterized by the model resolution matrix $\mathbf{R}_m = \mathbf{G}^\dagger \mathbf{G}$ in (10.16). In Chapter 4 we saw that this concept could easily be extended to Tikhonov regularization (4.20), by substituting the corresponding Tikhonov inverse matrix, $\mathbf{G}^\sharp$, for $\mathbf{G}^\dagger$ in (10.16).

We can recast (10.16) for nonlinear problems as

$$\mathbf{m} = G^{-1}(G(\mathbf{m}_{\text{true}})) \quad (10.17)$$

where G^{-1} and G are inverse and forward operators, respectively. However, the combined action of the forward and inverse operations is not representable as a matrix, such as $\mathbf{R}_m$ in (10.17), because the forward operator is a nonlinear function or algorithm, and the inverse operator is typically realized with an iterative method based on stepwise

linearization, such as GN or Occam's inversion. Furthermore, model resolution for a nonlinear inversion will not only depend on the physics, model parameterization, and data collection specifics, as was the case for linear problems, but may furthermore depend on the choice of starting model used in the solution algorithm, chosen convergence criteria, and possibly on the existence of multiple equally good solutions. Finally, as with linear methods, nonlinear model resolution will depend on the level and nature of the imposed regularization.

Because of these complexities, nonlinear resolution is typically analyzed using resolution tests. In seismic tomography, for example, it is common to evaluate the effects of nonideal resolution by generating noise-free synthetic data from a spike, checkerboard, or other test model using an identical source and receiver geometry as for the actual data in the problem of interest. A model is then recovered using the identical inverse methodology as was used for the actual data, and is compared to the test model to evaluate inversion artifacts. If there are specific features of the true model that will affect the resolution, such as a known strong velocity gradient with depth in seismic problems that significantly affects the curvature of the ray paths, those features should also be incorporated into the resolution test model. Because the problem is nonlinear, resolution analysis results will also potentially be dependent on test model amplitudes. A second resolution analysis strategy, sometimes referred to as a "squeeze" test, restricts the freedom of the model in regions that are suspected to be unnecessary or marginal for achieving an acceptable fit to the data. One example would be to modify the regularization constraints to strongly penalize model variations in the deeper part of a tomographic model to assess whether an acceptable data fit can still be achieved when structure is predominantly restricted to shallower depths.

Example 10.3

Revisiting [Example 10.1](#), we calculate noise-free synthetic data for a checkerboard velocity structure, using an identical starting model and ray-path end points. The checkerboard model ([Figure 10.9](#)) consists of an 8 by 8 node 2.9 km/s background model with alternating 10% variations for the 36 interior blocks. Inverting these data using the identical methodology as in [Example 10.1](#) for the same regularization constraints and range of regularization parameters, we obtain the suite of models shown in [Figure 10.10](#). For lower levels of regularization, the checkerboard is more apparent, while for higher levels, the horizontally varying velocity structure is substantially smoothed out. For the level of regularization chosen from the discrepancy principle for [Example 10.1](#), $\alpha \approx 37$, we note substantial smearing that makes it difficult to discern the full checkerboard pattern, indicating that structural variations of this character and spatial scale will be difficult to recover in some parts of the model without additional



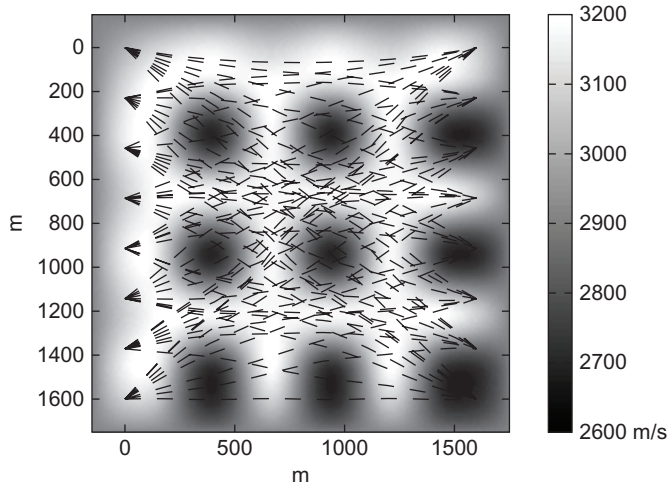


Figure 10.9 Checkerboard test model and ray paths for the cross-well tomography problem of Example 10.1.

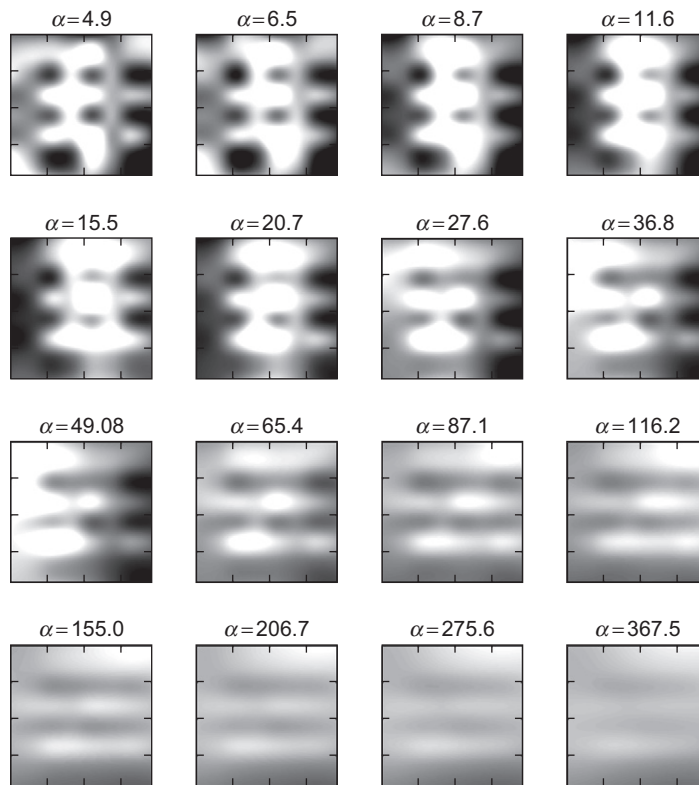


Figure 10.10 Suite of recovered models for a range of regularization parameter values (same as in Figure 10.2) for the checkerboard test model of Figure 10.9.

data. If we did not know the true model in [Example 10.1](#), this test would helpfully show that we would expect significant horizontal smearing for the inversion result shown in [Figure 10.4](#), and that the true model anomalies are thus not necessarily horizontally elongated.

10.4. EXERCISES

1. Show that, for a given value of α , the GN [\(10.7\)](#) and Occam's inversion model [\(10.14\)](#) update steps are mathematically equivalent.
2. Recall [Example 1.5](#), in which we had gravity anomaly observations above a density perturbation of variable depth $m(x)$, and fixed density $\Delta\rho$. Use Occam's inversion to solve an instance of this inverse problem. Consider a gravity anomaly along a 1-km section, with observations taken every 50 m, and density perturbation of $\Delta\rho = 200 \text{ kg/m}^3$ (0.2 g/cm^3). The perturbation is expected to be at a depth of roughly 200 m.



The MATLAB data file **gravprob.mat** contains a vector **x** of observation locations. Use the same coordinates for your discretization of the model. The vector **obs** contains the actual observations. Assume that the observations are accurate to about 1.0×10^{-12} .

- a. Derive a formula for the elements of the Jacobian.
 - b. Write MATLAB routines to compute the model predictions and the Jacobian for this problem.
 - c. Use the supplied implementation of Occam's inversion from [Example 10.2](#) to solve the inverse problem with second-order regularization.
 - d. Discuss your results. What features in the inverse solution appear to be real? What is the resolution of your solution? Were there any difficulties with local minimum points?
 - e. What would happen if the true density perturbation was instead at a depth of about 1000 m?
3. Apply the GN method with explicit regularization to the EM inversion problem by modifying the MATLAB code from [Example 10.2](#). Compare your solution with the solution obtained by Occam's inversion. Which method required more computational effort?
 4. Apply Occam's inversion to a cross-well, bent-ray tomography problem with identical geometry to [Example 10.1](#). Use the example subroutine **getj.m** to forward model the travel times, calculate the Jacobian, and implement second-order regularization. Travel-time data and subroutine control parameters are contained in the MATLAB data file **benddata.mat**. Start with the uniform 2900 m/s velocity 8 by 8 node initial



velocity model in **benddata.mat**, and assume independent and normally distributed data errors with $\sigma = 0.001$ ms.

Hint: A search range of α^2 between 10 and 10^5 is appropriate for this problem. MATLAB code for generating a second-order roughening matrix that approximates a two-dimensional Laplacian operator can be found in **makerough.m**.

10.5. NOTES AND FURTHER READING

In inverse problems with a large number of parameters, the most difficult computational problem is often computing derivatives of $\mathbf{G}(\mathbf{m})$ with respect to the model parameters, often referred to as **Fréchet derivatives** [107]. Computation of analytic formulas is commonly impractical. Finite difference estimates require computational effort, which increases with the number of parameters and may become impractical for large problems. A promising technique for calculating Fréchet derivatives is the adjoint approach [46, 154, 165], which is likely to see increasing use in large problems as forward modeling capabilities advance. An alternative approach involves using the discretized differential equation as a set of constraints to be added to the nonlinear least squares problem [15].

For large-scale problems, it may be impractical to use direct factorization to solve the systems of equations (10.7) or (10.14) involved in computing the GN or Occam step. One approach in this case is to use an iterative method such as conjugate gradients to solve the linear systems of equations [58]. The conjugate gradient method can also be extended to minimize the nonlinear objective function directly [58, 142].